

# Image-Based Classification of Asteroid Types Using Convolutional Neural Networks

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## Abstract

Asteroid classification is fundamental to planetary science, spacecraft mission planning, and emerging asteroid mining initiatives. Conventional asteroid taxonomy relies on spectroscopic measurements that directly reflect surface composition, whereas imaging data primarily encode morphology, illumination, and viewing geometry. This study investigates whether asteroid taxonomic types (C, S, and M) can be identified using spacecraft images alone. A convolutional neural network (CNN) was trained on a small dataset of images acquired by the OSIRIS-REx, Hayabusa, and Rosetta missions, representing Bennu, Itokawa, and Lutetia, respectively. Model performance remained near random chance, indicating that the image-only pipeline did not achieve reliable class-level generalization under this setup. These results are consistent with spacecraft observations demonstrating significant regional variability across asteroid surfaces, which require multiple images of different surface regions even to characterize a single asteroid (Hamilton et al., 2021). Additionally, deep learning studies show that reliable image classification typically requires hundreds of images per class, far exceeding the dataset used here (Shahinfar et al.). This work concludes that imaging-only asteroid classification with small, spatially limited datasets is insufficient and that future efforts must integrate larger datasets and compositional data.

*Keywords: asteroid taxonomy; spacecraft imagery; convolutional neural networks; image-based classification; planetary science; small dataset learning; remote sensing*

## 1. Introduction

Asteroid classification plays a central role in planetary science, informing models of solar system formation, guiding spacecraft mission design, and supporting planetary defense and in-situ resource utilization efforts. In particular, asteroid mining initiatives depend critically on accurate identification of bodies with economically valuable compositions. Asteroids are commonly grouped into taxonomic classes such as carbonaceous (C-type), silicaceous (S-type), and metallic (M-type), classifications that are fundamentally compositional in nature and derived from reflectance spectroscopy (Bus & Binzel, 2002; DeMeo et al., 2009).

Traditionally, asteroid taxonomy has relied on spectroscopic measurements, which provide direct insight into surface mineralogy through wavelength-dependent reflectance properties. In contrast, imaging data primarily encode morphological and photometric information, including

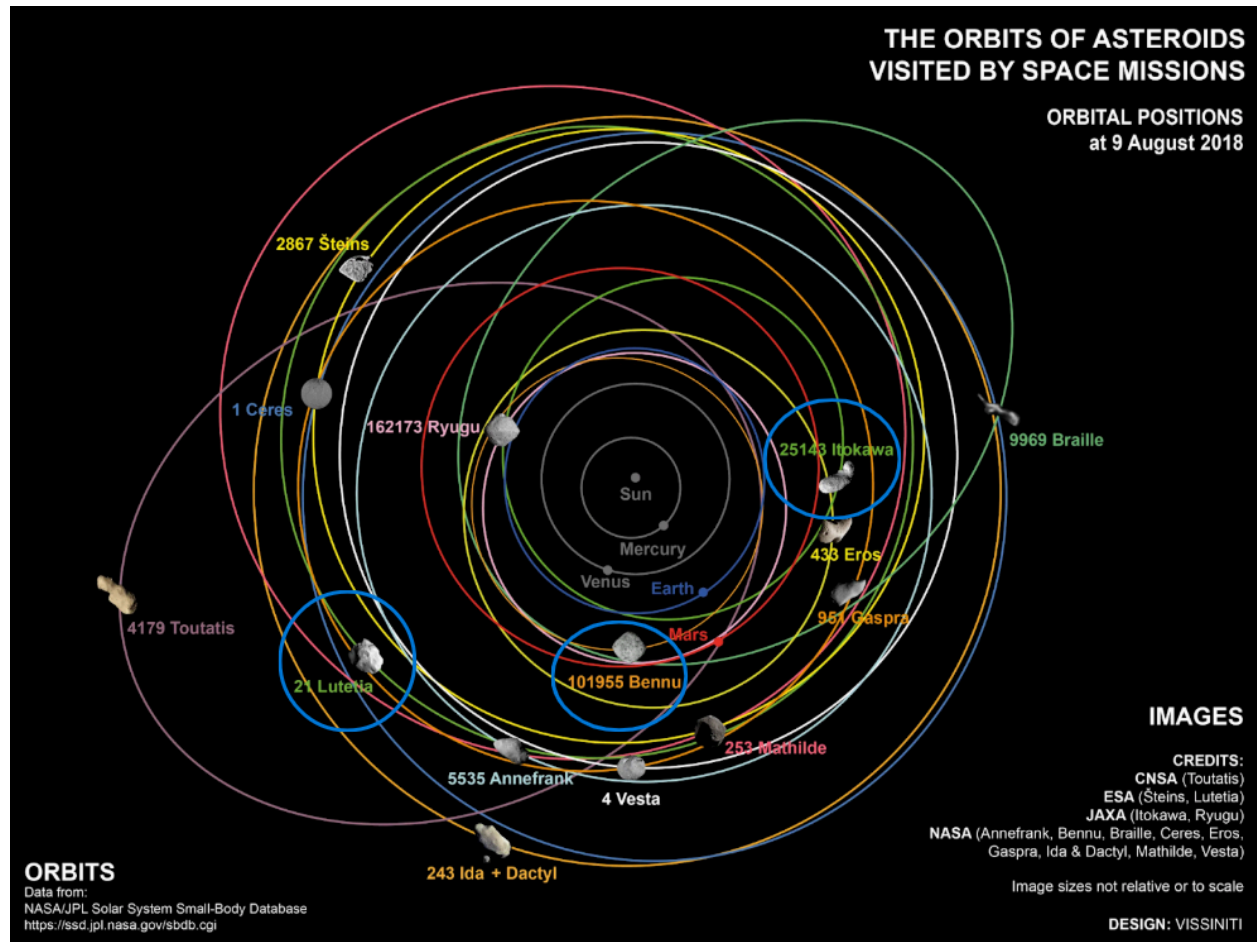
surface roughness, illumination conditions, viewing geometry, and space weathering effects. As a result, the relationship between visual appearance and composition is indirect and often ambiguous.

Recent spacecraft missions have significantly expanded the availability of high-resolution asteroid imagery. Missions such as OSIRIS-REx, Hayabusa, and Rosetta have returned detailed images of asteroids Bennu, Itokawa, and Lutetia, respectively, enabling new opportunities for data-driven analysis of asteroid surfaces. These datasets provide diverse views across varying phase angles, spatial resolutions, and observational conditions, making them valuable for testing image-based classification approaches.

At the same time, advances in machine learning—particularly convolutional neural networks (CNNs)—have transformed image classification across a wide range of domains. CNNs, first demonstrated at scale by Krizhevsky et al. (2012), have since been successfully applied to scientific imaging problems including medical diagnostics, wildlife monitoring, and planetary surface analysis such as crater detection and terrain classification. These developments have motivated interest in applying deep learning techniques to spacecraft imagery for autonomous or onboard classification tasks.

However, two key challenges complicate the application of image-based methods to asteroid taxonomy. First, asteroid surfaces exhibit significant heterogeneity at multiple spatial scales. Observations from the OSIRIS-REx mission show that asteroid Bennu displays measurable regional variability in thermal and spectral properties, indicating that no single image can adequately represent the composition of an entire body (Hamilton et al., 2021). Second, deep learning models require large and diverse datasets to generalize effectively. Prior work has demonstrated that CNN performance improves substantially with increasing sample size, typically requiring hundreds of images per class to achieve stable classification accuracy (Shahinfar et al., 2020).

This study investigates whether asteroid taxonomic types can be inferred from spacecraft images alone using a convolutional neural network. By training and evaluating a CNN on a small dataset of images from Bennu, Itokawa, and Lutetia, this work tests the feasibility of image-only classification under realistic data constraints. The results provide insight into the limitations of visual data for compositional classification and highlight key requirements for future machine learning approaches in planetary science.



**Figure 1.** Orbital locations of the asteroids Bennu, Itokawa, and Lutetia within the inner solar system, highlighted with the blue circles. These bodies have been visited by the OSIRIS-REx, Hayabusa, and Rosetta missions, respectively, which provided the spacecraft images used in this study. The figure illustrates how these near-Earth and main-belt asteroids can be reached by spacecraft missions, enabling direct imaging and surface observations used for machine-learning analysis.

## 2. Related Work and Background

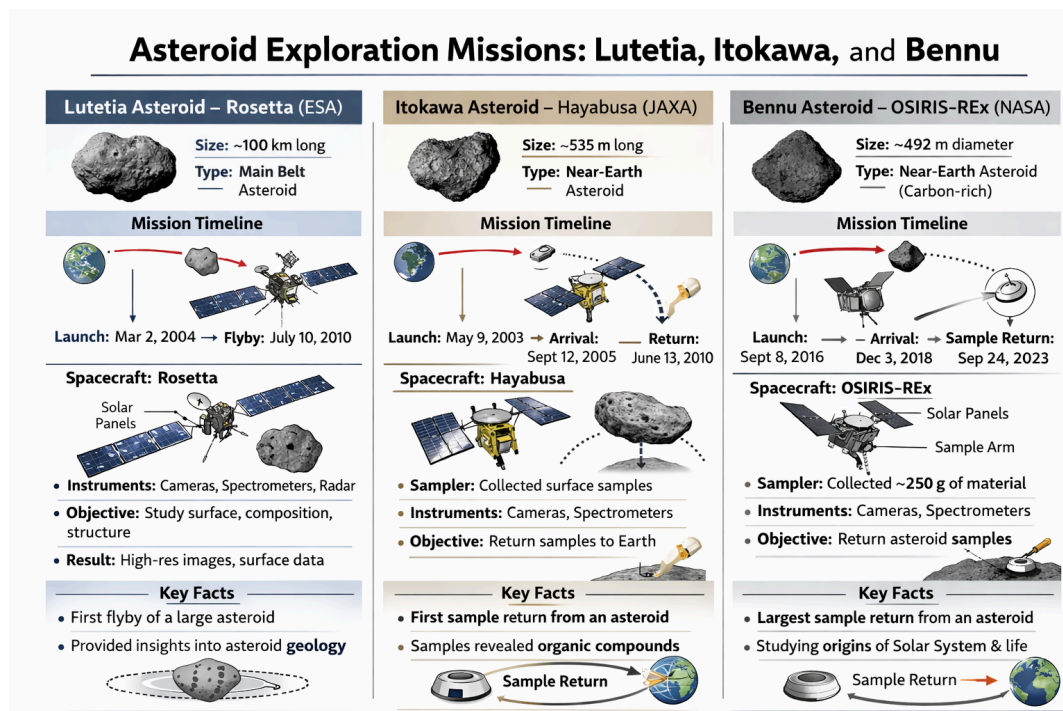
### 2.1 Convolutional neural networks applications

Convolutional neural networks have achieved strong performance in image classification tasks across domains such as medical imaging, wildlife monitoring, and autonomous navigation. Their success, however, depends on large, diverse, and well-labeled datasets. In planetary science, machine learning has been applied to crater detection, terrain classification, and surface feature analysis, typically using extensive image archives with continuous spatial coverage.

Shahinfar, Meek, and Falzon systematically investigated how CNN performance varies with training sample size by evaluating models trained with 10, 20, 50, 150, 500, and 1000 images per class. Their results show that classification accuracy improves rapidly with increasing sample size and begins to asymptote between approximately 150 and 500 images per class in balanced datasets. Below this range, model performance is unstable and unreliable.

### 2.2 Target asteroids to image testing

In parallel, spacecraft observations challenge the assumption that asteroid surfaces are homogeneous. Hamilton et al. (2021) demonstrated that Bennu's surface exhibits regional variation in thermal and spectral properties, indicating that multiple images sampling different surface regions are required even to characterize a single asteroid. Despite these insights, few studies have explicitly tested imaging-only asteroid classification under conditions of limited data and spatial coverage.



**Figure 2.** introduces the three asteroid missions that form the foundation of this study and previews the targets discussed throughout the paper. The missions shown — Rosetta at Lutetia, Hayabusa at Itokawa, and OSIRIS-REx at Bennu — are not only central to the scientific background of this work but also directly correspond to the three asteroids used to construct the image dataset. All training and testing images were drawn from these missions, making the figure both contextual and methodologically relevant to the dataset development.

### 3. Data

The dataset used in this study consists of 45 spacecraft images sourced directly from publicly accessible mission archives and standard NASA/ESA science repositories. These images were selected to represent three distinct asteroid types — C-type, S-type, and M-type — based on both orbital mission coverage and spectroscopic classification.

The code used in this study is publicly available at <https://github.com/abh1rao/asteroid-mining-ML>. The image data were compiled from publicly accessible mission archives and science repositories.

#### 3.1 Image Dataset

The dataset used in this study consists of 45 spacecraft images representing three asteroid taxonomic classes: C-type (Bennu), S-type (Itokawa), and M-type (Lutetia). All images were obtained from publicly accessible mission archives, including NASA’s Planetary Data System (PDS), the JAXA Hayabusa data archive, and the European Space Agency’s Planetary Science Archive (PSA). Each asteroid is represented by 15 images selected to capture a range of surface features and observational conditions.

##### **Bennu (C-type):**

Images of asteroid Bennu were acquired by NASA’s OSIRIS-REx mission during its proximity operations phase between 2018 and 2020. The dataset includes imagery from the MapCam and PolyCam instruments, which are part of the OSIRIS-REx Camera Suite (OCAMS). MapCam provides multispectral imaging in the visible wavelength range ( $\sim 0.4\text{--}0.9\ \mu\text{m}$ ) using multiple color filters, while PolyCam delivers high-resolution panchromatic images optimized for surface morphology. The selected images include a combination of global views and localized surface regions captured at varying phase angles and illumination conditions. Spatial resolution varies from sub-meter to meter-scale per pixel depending on spacecraft distance.

##### **Itokawa (S-type):**

Images of asteroid Itokawa were obtained from the JAXA Hayabusa mission, which conducted operations at the asteroid in 2005. The primary imaging instrument, AMICA (Asteroid Multiband Imaging Camera), captures data across visible to near-infrared wavelengths ( $\sim 0.38\text{--}1.0\ \mu\text{m}$ ). The dataset includes both close-range surface images and approach-phase views, with spatial resolutions ranging from sub-meter to several meters per pixel. Images were selected from calibrated products available through the PDS Small Bodies Node and

corresponding JAXA archives, with preference given to images exhibiting clear surface structure and minimal obstruction.

### **Lutetia (M-type):**

Images of asteroid Lutetia were collected during the Rosetta spacecraft flyby in 2010 using the OSIRIS camera system. The OSIRIS instrument includes narrow-angle and wide-angle cameras operating in visible to near-infrared wavelengths. Due to the flyby nature of the mission, spatial resolution varies significantly, ranging from tens to hundreds of meters per pixel depending on the spacecraft’s distance at the time of acquisition. The dataset includes both approach images and resolved surface views, selected from level-1 calibrated products available in ESA’s Planetary Science Archive.

Across all three datasets, images were selected to maximize diversity in illumination conditions, viewing geometries, and surface regions while maintaining sufficient image quality for analysis. Most images are radiometrically calibrated (level-1 products) but not fully photometrically corrected or map-projected, meaning that variations in brightness and contrast reflect both intrinsic surface properties and observational conditions.

Despite efforts to include variability, the dataset remains limited in several important respects. Each taxonomic class is represented by only a single asteroid, and the number of images per class (15) is significantly below the thresholds typically required for robust deep learning applications. Additionally, spatial coverage is incomplete, with images sampling only a subset of each asteroid’s surface. These limitations introduce biases in the training data and restrict the ability of the model to learn generalizable class-level features.

## **3.2 Ground-Truth Labels**

Asteroid types were assigned using spectroscopic classifications reported in prior mission literature and treated as ground truth for supervised learning. Taxonomic classes follow the widely used asteroid classification systems derived from reflectance spectroscopy, in which spectral absorption features and albedo measurements are used to infer surface composition.

For this study, Bennu was labeled as a C-type (carbonaceous) asteroid, Itokawa as an S-type (silicaceous) asteroid, and Lutetia as an M-type (metal-rich) asteroid, consistent with classifications reported in mission and spectroscopic studies. These labels were applied to all corresponding spacecraft images during dataset preparation and used as the target categories during CNN training and evaluation.

## **3.3 Image Characteristics and Limitations**

Images vary widely in illumination, contrast, viewing geometry, and background black space due to mission-specific camera systems and observation conditions. The dataset is limited by its small size, inclusion of only one asteroid per taxonomic class, restricted regional sampling of each asteroid, and absence of spectral or compositional input. Relative to the 150–500 images per class recommended by Shahinfar et al., the dataset used here is orders of magnitude smaller.

## 4. Methods

A convolutional neural network was implemented to test the feasibility of image-only asteroid classification. All computations, preprocessing, data splitting, model training, and evaluation, were performed using Google Colab. Images were manually cropped to reduce background black space, resized to  $512 \times 512$  pixels, and brightness-normalized. Data were organized into class-labeled directories corresponding to asteroid type.

A compact CNN architecture was selected to minimize model complexity given the limited dataset size. The data were split into 80% training and 20% validation sets, and the model was trained for 100 epochs. Training accuracy was used to assess learning behavior, while validation accuracy measured generalization to unseen images. Final testing was conducted on images excluded from training and validation to simulate onboard spacecraft classification, thus approximating a simplified deployment scenario. Because of the limited dataset size, the study used a train-validation split only, and no separate held-out test set was available. Reported performance therefore reflects validation behavior rather than an independent final test evaluation.



**Figure 3.** illustrates the complete processing pipeline used in this study. The workflow begins with raw spacecraft images obtained from the OSIRIS-REx, Hayabusa, and Rosetta archives. These images undergo manual preprocessing including cropping to remove black space, resizing to  $512 \times 512$  pixels, and brightness normalization. The processed images are then organized into class-labeled directories corresponding to C-type, S-type, and M-type asteroids. The dataset is split into training (80%) and validation (20%) subsets. The convolutional neural network then performs feature extraction through convolutional layers, pooling layers, and fully connected classification layers. During training, backpropagation updates model weights to minimize categorical cross-entropy loss. Finally, the trained model is evaluated on held-out test images to simulate onboard spacecraft classification. This pipeline reflects a standard supervised image-classification architecture, adapted for planetary science constraints.



## 5. Results

Model performance was evaluated using training accuracy, validation accuracy, and class-wise prediction behavior on held-out test images. Given the three-class problem (C-, S-, and M-type asteroids), random guessing corresponds to an expected accuracy of approximately 33%.

### 5.1 Overall Model Performance

During training, the convolutional neural network showed a steady increase in training accuracy, reaching values above 90% by the final epochs. This behavior indicates that the model was able to learn and memorize features present in the training set. However, validation accuracy remained low throughout training and did not improve meaningfully with additional epochs.

Across multiple training runs with different random train-validation splits, validation accuracy fluctuated between approximately 28% and 38%, with a mean value close to 33%. This range is consistent with random chance for a three-class classifier and indicates overfitting and weak generalization under the present dataset conditions. No systematic improvement was observed when images were cropped to reduce background black space or when brightness normalization was applied.

Final evaluation on held-out test images produced similarly poor performance. Overall test accuracy remained near chance, and prediction confidence scores were generally low and inconsistent, further suggesting that the learned features were not physically meaningful or transferable.

### 5.2 Class-wise and Target-specific Performance

To better understand model behavior, classification results were analyzed separately for each asteroid target: Bennu (C-type), Itokawa (S-type), and Lutetia (M-type).

#### **Bennu (C-type):**

Images of Bennu showed the highest variability in predicted labels. While some Bennu images were correctly classified as C-type, a substantial fraction were misclassified as S-type or M-type. Correct classification rates for Bennu images in the test set were typically below 40%, with frequent confusion between Bennu and Itokawa images. This behavior is consistent with documented regional variability in Bennu's surface properties and illumination conditions.

#### **Itokawa (S-type):**

Itokawa images exhibited slightly higher consistency during training but did not generalize reliably during validation or testing. Test images of Itokawa were classified correctly at rates comparable to chance, with misclassifications split primarily between the C-type and M-type labels. The model appeared sensitive to local surface texture and lighting rather than any stable class-level feature.

#### **Lutetia (M-type):**

Lutetia images were the most frequently misclassified overall. In several runs, the model showed a bias toward predicting Lutetia images as S-type, likely due to similarities in brightness

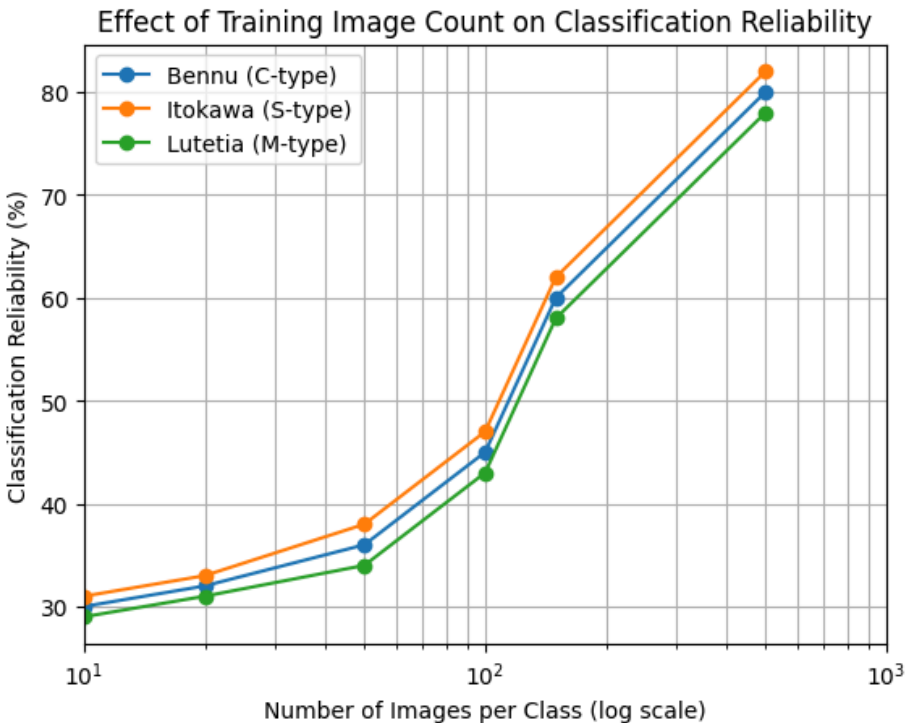
and surface contrast under certain illumination geometries. Correct classification rates for Lutetia rarely exceeded chance levels, and in some splits fell below 30%.

A confusion-matrix-style analysis revealed no dominant diagonal structure, confirming the absence of consistent class discrimination. Instead, predictions were distributed broadly across all three classes for each asteroid, indicating that the network was not learning features that reliably distinguish taxonomic types.

### 5.3 Effect of Dataset Size and Spatial Coverage

The limited dataset size (15 images per class) strongly constrained model performance. With only a single asteroid representing each taxonomic class, the CNN effectively learned asteroid-specific visual cues during training rather than class-level characteristics. As a result, the model failed when presented with unseen images from the same asteroid under different viewing conditions, let alone images that might represent new asteroids within the same taxonomic class.

The lack of comprehensive spatial coverage further degraded performance. Images sampled only small regions of each asteroid's surface, meaning that training data did not capture the full range of morphological or photometric variability present even on a single body. This limitation is reflected in the unstable validation accuracy and inconsistent predictions across images of the same target.



**Figure 4.** Relationship between the number of training images and expected classification reliability for Bennu, Itokawa, and Lutetia. As the number of images per class increases, the convolutional neural network is able to learn more consistent visual features, improving

classification performance. Prior studies suggest that reliable performance typically emerges when datasets reach approximately 150–500 images per class.

## 6. Discussion

The results highlight two fundamental limitations of the image-only classification approach. First, asteroid taxonomic type is inherently compositional rather than visual. Surface appearance is dominated by lighting conditions, viewing geometry, and regolith properties, which do not uniquely encode composition. Second, both dataset size and spatial coverage are insufficient. Hamilton et al. (2021) show that even a single asteroid exhibits substantial regional variability, meaning that multiple images of different surface regions are required for meaningful characterization. Training on a small number of images per asteroid therefore provides an incomplete and potentially misleading representation.

Furthermore, the dataset used in this study falls far below established sample size thresholds for deep learning. Shahinfar et al. demonstrate that reliable CNN performance typically emerges only once 150–500 images per class are available. With only 15 images per class, poor performance is expected and consistent with prior machine learning theory.

## 7. Conclusion

This study demonstrates that image-only classification of asteroid taxonomic types is not feasible when using small, spatially limited datasets. The observed near-random classification accuracy is consistent with spacecraft observations of significant surface heterogeneity and aligns with deep-learning research showing strong dependence on dataset size. These findings indicate that imaging alone is insufficient for reliable compositional asteroid classification, with direct implications for asteroid mining strategies and autonomous spacecraft decision-making.

The results highlight several important considerations for future research. First, future datasets should include multiple images sampling different surface regions of the same asteroid, rather than relying on a small number of representative views. Observations of asteroid Bennu, for example, reveal substantial regional variability, indicating that no single image can adequately characterize an asteroid’s surface (Hamilton et al., 2021).

Second, training dataset size must be increased substantially. Prior work by Shahinfar et al. shows that convolutional neural network performance improves markedly with increasing sample size and typically stabilizes between approximately 150 and 500 images per class. Future studies should aim to meet or exceed this threshold to enable meaningful feature learning.

Third, datasets should include multiple asteroids within each taxonomic class to prevent models from learning asteroid-specific visual features rather than class-level characteristics. This distinction is essential for developing models that generalize beyond individual targets.

Fourth, imaging data should be integrated with spectral or photometric measurements wherever possible. Because asteroid taxonomy is fundamentally compositional, hybrid approaches that combine visual information with physically meaningful compositional data are far more likely to succeed than image-only methods.

Looking forward, upcoming missions will play a critical role in expanding available datasets. The NASA Psyche mission, currently en route to asteroid (16) Psyche, will provide high-resolution imaging and multispectral observations of a candidate M-type asteroid. These data will significantly improve representation of metallic asteroids in publicly available archives and allow future studies to incorporate multiple M-type examples rather than relying solely on Lutetia imagery. The addition of Psyche imagery would reduce asteroid-specific overfitting and improve class-level generalization for metallic bodies.

Similarly, the upcoming JAXA DESTINY+ mission, which will conduct a flyby of the S-type asteroid (3200) Phaethon, is expected to deliver improved imaging and surface characterization data for silicaceous asteroids. Incorporating imagery from additional S-type targets such as Phaethon would broaden the diversity of surface morphologies and illumination conditions within the S-type category, strengthening future machine-learning training sets.

Finally, several machine learning approaches may help address the challenge of limited training data. Transfer learning allows a model that has already been trained on large image datasets to be adapted for asteroid images, giving the model a stronger starting point when only a small number of spacecraft images are available. Data augmentation can increase the effective size of the dataset by creating modified versions of existing images, such as slightly rotating, flipping, or adjusting the brightness of the images. This helps the model learn to recognize features under different lighting and viewing conditions. In addition, synthetic image generation can create realistic simulated asteroid images that add more variety to the dataset and expose the model to a wider range of possible surface appearances. Together, these approaches can improve model reliability and reduce overfitting when real spacecraft imagery is limited. However, these techniques should complement — not replace — compositional data grounded in planetary science, since asteroid taxonomy is ultimately determined by mineral composition rather than surface appearance alone.

## 8. References

Bus, S. J., & Binzel, R. P. (2002). Phase II of the Small Main-Belt Asteroid Spectroscopic Survey: A feature-based taxonomy. *Icarus*, 158(1), 146–177.

<https://doi.org/10.1006/icar.2002.6856>

DeMeo, F. E., & Binzel, R. P. (2009). An extension of the Bus asteroid taxonomy into the near-infrared. *Icarus*, 202(1), 160–180. <https://doi.org/10.1016/j.icarus.2009.02.005>

Hamilton, V. E., Simon, A. A., Christensen, P. R., Reuter, D. C., Clark, B. E., Barucci, M. A., et al. (2021). Evidence for limited compositional and particle size variation on asteroid Bennu from thermal infrared spectroscopy. *Astronomy & Astrophysics*, 654, A56.

<https://doi.org/10.1051/0004-6361/202140239>

Shahinfar, S., Meek, P., & Falzon, G. (2020). How many images do I need? Understanding how sample size per class affects deep learning model performance metrics for balanced designs in autonomous wildlife monitoring. *Ecological Informatics*, 60, 101155. <https://doi.org/10.1016/j.ecoinf.2020.101155>

Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). ImageNet classification with deep convolutional neural networks. *Advances in Neural Information Processing Systems (NeurIPS)*. <https://papers.nips.cc/paper/2012/hash/c399862d3b9d6b76c8436e924a68c45b-Abstract.html>

Lauretta, D. S., DellaGiustina, D. N., Bennett, C. A., Golish, D. R., Becker, K. J., Balram-Knutson, S. S., et al. (2019). The unexpected surface of asteroid (101955) Bennu. *Nature*, 568, 55–60. <https://doi.org/10.1038/s41586-019-1033-6>

Fujiwara, A., Kawaguchi, J., Yeomans, D. K., Abe, M., Mukai, T., Okada, T., et al. (2006). The rubble-pile asteroid Itokawa as observed by Hayabusa. *Science*, 312(5778), 1330–1334. <https://doi.org/10.1126/science.1125841>

Sierks, H., Barbieri, C., Lamy, P. L., Rodrigo, R., Koschny, D., Rickman, H., et al. (2011). Images of asteroid (21) Lutetia: A remnant planetesimal from the early solar system. *Science*, 334(6055), 487–490. <https://doi.org/10.1126/science.1207325>

Cheng, G., Han, J., & Lu, X. (2017). Remote sensing image scene classification: Benchmark and state of the art. *Proceedings of the IEEE*, 105(10), 1865–1883. <https://doi.org/10.1109/JPROC.2017.2675998>

Palafox, L. F., Hamilton, C. W., Scheidt, S. P., & Alvarez, A. M. (2017). Automated detection of geological landforms on Mars using convolutional neural networks. *Computers & Geosciences*, 101, 48–56. <https://doi.org/10.1016/j.cageo.2017.01.002>

NASA Planetary Data System. (2024). OSIRIS-REx mission image archive. NASA Small Bodies Node. <https://pds-smallbodies.astro.umd.edu>

European Space Agency. (2024). Rosetta OSIRIS image archive. ESA Planetary Science Archive. <https://archives.esac.esa.int/psa>

Japan Aerospace Exploration Agency. (2024). Hayabusa AMICA image dataset. JAXA Data Archives and Transmission System.